

00 · Foundations — the words the cookbook assumes

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The notebook that comes **before** everything else. It builds the core vocabulary — potential outcomes, counterfactuals, confounding, randomization, identification, the do-operator, DAGs, ATE vs CATE — from nothing, with one marketing example, and ends with a worked **estimator ladder**: naive $\rightarrow$ regression-adjust $\rightarrow$ IPW $\rightarrow$ doubly-robust, each recovering a known effect from confounded data.

No new theory here — it mirrors `docs/causal_inference_primer.pdf`. Skim it once and every later notebook reads as a *named tool* rather than new vocabulary.

1.1 1 · Treatment, unit, outcome — and the counterfactual

A **treatment** is any action whose effect we want (send a discount, show an ad, change a price). The **unit** receives it (customer, store, region); the **outcome** is what we measure after.

For binary $T \in \{0, 1\}$ each unit has **two potential outcomes**: $Y(1)$ if treated, $Y(0)$ if not. We only observe **one** — $Y = TY(1) + (1 - T)Y(0)$ — so the individual effect $Y(1) - Y(0)$ is never seen. This **fundamental problem of causal inference** is a *missing-data* problem; everything downstream is principled imputation of the half of the table we didn't get to see.

	customer	$Y(0)$ not emailed	$Y(1)$ emailed	individual effect	emailed?	\
0	Anna	30	80	50	yes	
1	Ben	55	60	5	no	
2	Cara	20	22	2	yes	

	$\rightarrow$ OBSERVED	$\rightarrow$ MISSING (counterfactual)
0	80	30
1	55	60
2	22	20

1.2 2 · From individuals to averages, and the enemy: confounding

Decisions run on **averages**: the **Average Treatment Effect** $ATE = \mathbb{E}[Y(1) - Y(0)]$. The whole difficulty collapses onto one word — **comparable**: to estimate $\mathbb{E}[Y(0)]$ for the treated we need a control group *like them in every way except the treatment*.

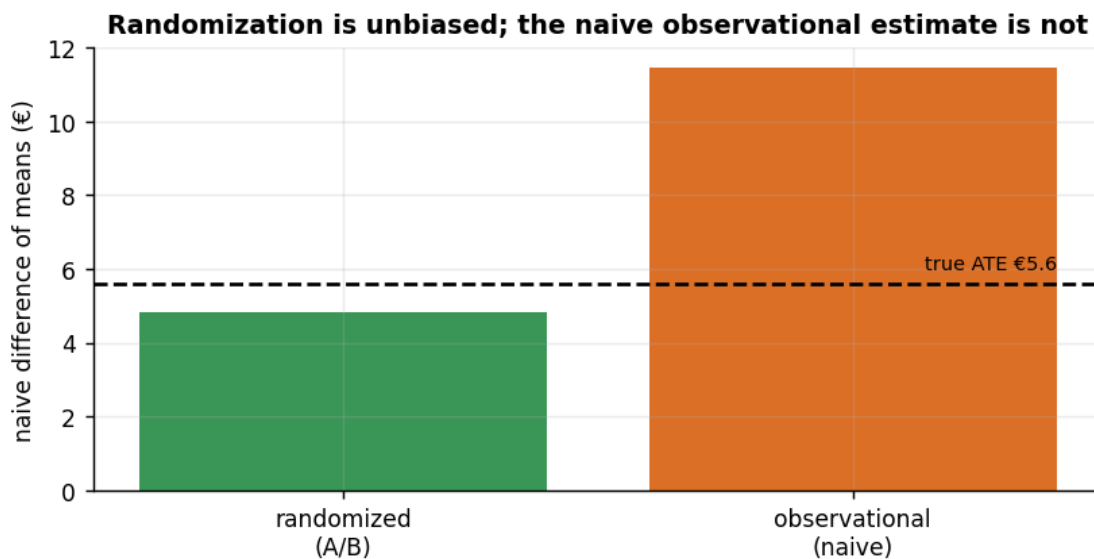
Confounding breaks comparability. If we email our *loyal* customers and compare to *casual* controls, loyal customers spend more anyway — “loyalty” drove both who got emailed and how much they spend. A **confounder** is a common cause of treatment and outcome; the naive “treated —

control” difference then blames the treatment for a gap that was really about *who ended up in each group*.

1.3 3 · The gold standard, and life without it

Randomization (an A/B test) annihilates confounding by design: a coin flip makes treatment independent of every confounder, measured or not, so $ATE = \langle Y \mid T=1 \rangle - \langle Y \mid T=0 \rangle$ — a plain difference of means. We can *see* this: the same simulator run in **randomized** vs **observational** mode gives the same true effect, but only the randomized naive estimate is unbiased.

true ATE €5.6 · randomized naive €4.8 (unbiased) · observational naive €11.4 (confounded)



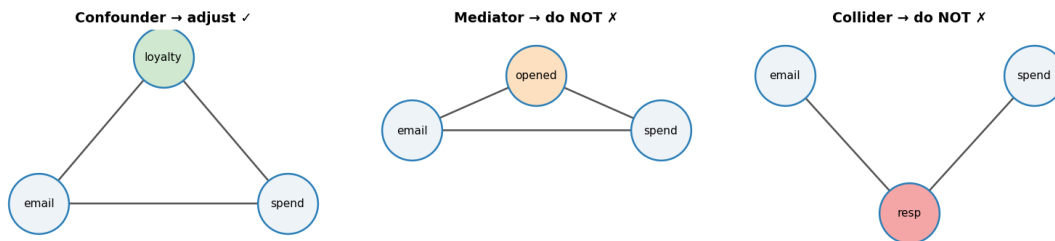
Without randomization we must **identify** the effect using assumptions:

1. **Unconfoundedness** $\{Y(0), Y(1)\} \perp T \mid X$ — within units alike on measured X , treatment is as-good-as-random. (*The big, untestable one.*)
2. **Positivity / overlap** $0 < e(x) < 1$, $e(x) = P(T=1 \mid X=x)$ the **propensity score** — every kind of unit could have gone either way.
3. **SUTVA** — no interference, one version of treatment.

Under these, the **adjustment (g-)formula** compares like-with-like *within* strata of X and averages back up: $ATE = \mathbb{E}_X[\mathbb{E}[Y \mid T=1, X] - \mathbb{E}[Y \mid T=0, X]]$.

1.4 4 · Seeing vs doing (the do-operator), DAGs, colliders

Observing $X = x$ (units where X happened to be x , carrying all the reasons it got there) differs from **doing** $do(X = x)$ (reaching in and *setting* it). Causal effects are about *doing*. A **DAG** draws each variable as a node and each direct cause as an arrow, and tells you exactly what to control for:



- **Backdoor path** (arrow *into* T, e.g. $T \leftarrow \text{loyalty} \rightarrow Y$): block it by conditioning — the minimal blocking set is the X in the adjustment formula.
- **Mediator** ($T \rightarrow M \rightarrow Y$): controlling for M removes part of the effect you want.
- **Collider** ($T \rightarrow C \leftarrow Y$): conditioning on C *manufactures* a spurious association. Notebook 05 is the whole “what to control for” story.

1.5 5 · One effect for everyone, or a different effect for each?

- **ATE** — one number (“did the program lift revenue overall?”): **program evaluation / incrementality** (notebooks 06–11).
- **CATE** $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X=x]$ — the effect as a function of *who*: **heterogeneous treatment effects** (notebooks 01–04). If τ were identical for everyone, there’d be no one to select — targeting works *because* $\tau(x)$ varies.

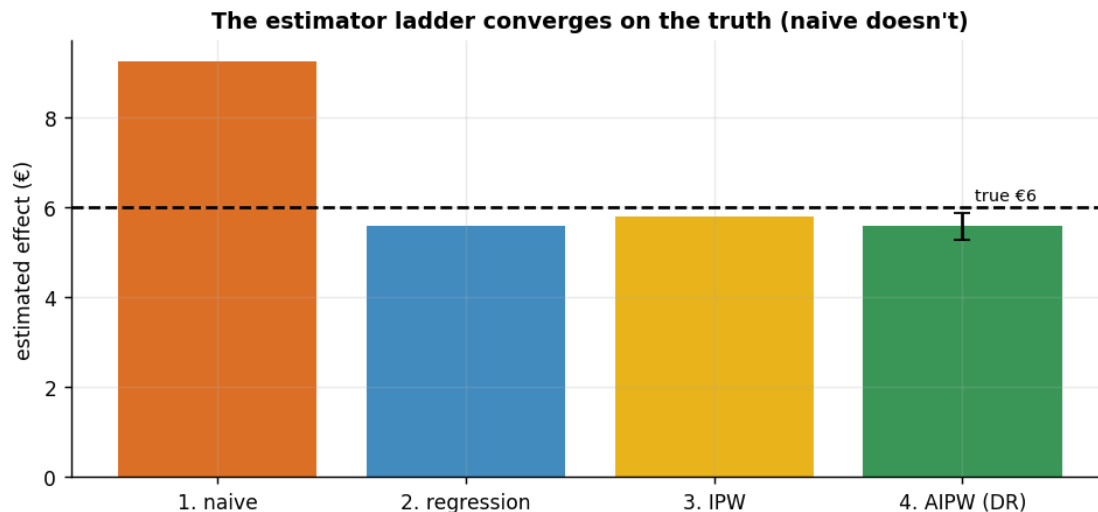
(Sometimes a method only recovers the effect for a sub-population — e.g. the compliers an instrument nudges. That restricted quantity is a **LATE**; notebook 11.)

1.6 6 · The estimator ladder — four ways to recover one effect

Where the machinery enters. On the **confounded** data (true effect €6), we climb a ladder of estimators, each leaning on different assumptions, and watch them converge on the truth as we do the adjustment right:

1. **Naïve** difference of means — ignores X , biased up.
2. **Regression adjustment** — model $\mathbb{E}[Y \mid T, X]$, average the contrast (right if the outcome model is right).
3. **IPW** — reweight by the inverse propensity so treated/control look alike (right if the propensity model is right).
4. **AIPW (doubly-robust)** — combine both; consistent if *either* model is right, with a credible interval.

1. naïve	€ 9.26	(true €6.0)
2. regression	€ 5.60	(true €6.0)
3. IPW	€ 5.80	(true €6.0)
4. AIPW (DR)	€ 5.59	(true €6.0)



The discipline: **Bayes/ML sharpens estimation, never identification.** A tight, beautiful posterior computed under a false unconfoundedness assumption is *confidently wrong*. Priors don't fix broken causal arguments — the graph does. Every notebook keeps identification (this ladder's assumptions) separate from estimation (which rung you climb).

1.6.1 The eight words to walk in with

Treatment · **counterfactual** / **potential outcome** (the branch you didn't observe) · **confounder** (common cause of treatment and outcome) · **randomization** (kills confounding by design) · **identification** (the assumptions that make an effect recoverable) · **do-operator** (forcing vs observing) · **backdoor** / **collider** (what to control for, and what never to) · **ATE vs CATE** (one effect for all vs a different effect per customer).

Next: 01 · [uplift targeting](#) (Anchor A, HTE) and 07 · [geo-lift synthetic control](#) (Anchor B, program evaluation) — then the rest of the cookbook.